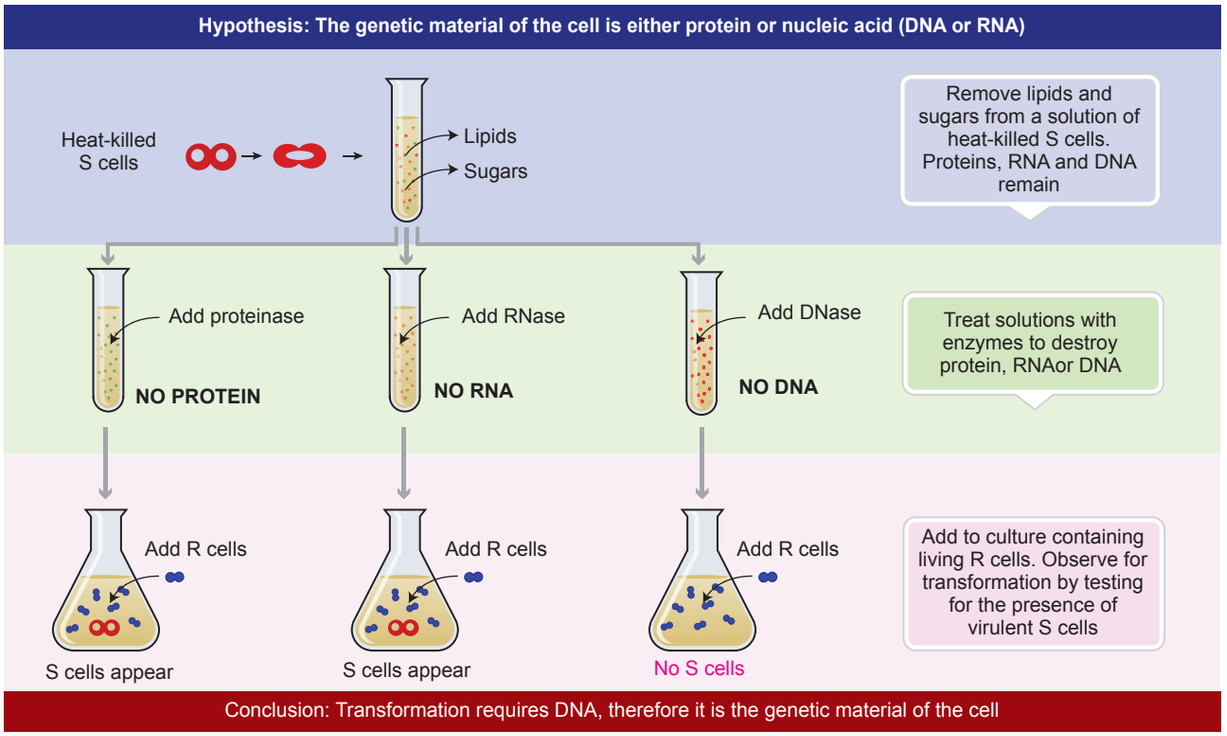
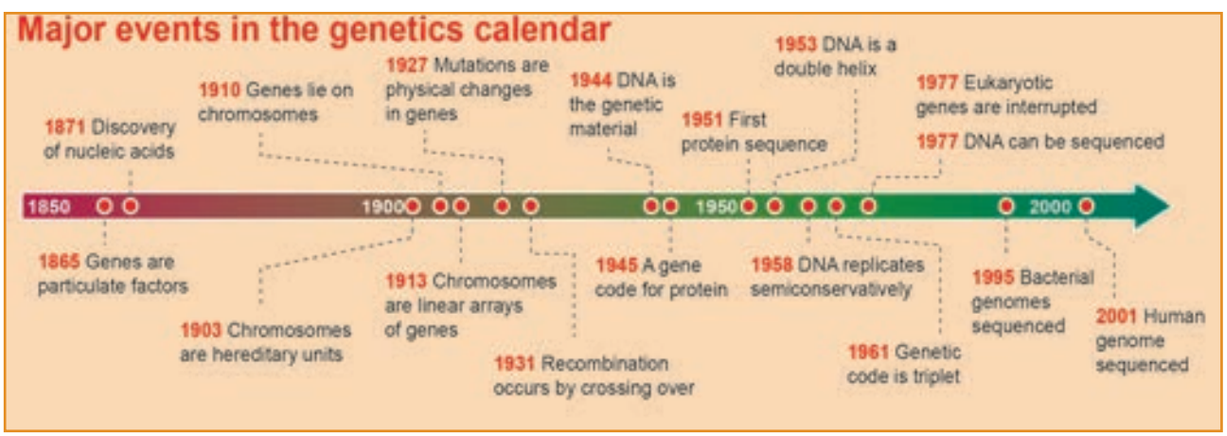


Fig. 5.1 Transformation experiment of Avery *et. al.*, (1944)

to identify the nature of the transforming substance responsible for converting a non-virulent strain into virulent strain. They observed that the DNA, RNA and proteins isolated from the heat-killed S-strain when added to R-strain changed their surface character from rough to smooth and also made them pathogenic **(Fig. 5.1)**. But when the extract was treated with **DNase** (an enzyme which destroys DNA) the transforming ability was lost. **RNase** (an enzyme which destroys RNA) and **proteases** (an enzyme which destroys protein) did not affect the transformation. Digestion with DNase inhibited transformation suggesting that the DNA caused the transformation. These experiments suggested that DNA and not proteins is the genetic material. The phenomenon, by which DNA isolated from one type of cell (S – strain), when introduced into another type (R-strain), is able to retain some of the properties of the S - strain is referred to as **transformation**.



TN_GOVT_XII_Bio_Zoology_EM_chapter 5.indd 60 20-03-2020 10.33.21 AM